

Discrete Mathematics Final Exam (Spring 2016)

No :

Name:

- (30P) Tic-tac-toe is a game for two players (X and O) who take turns marking the spaces in a 3×3 grid. The player who succeeds in placing three of their marks in a horizontal, vertical, or diagonal row wins the game. When the game is as below, it is X's turn. For this situation, please go on to draw the game tree and comment α - β pruning on this tree.

O	O	X
	X	
O	X	

Object function produces 1 for X, -1 for O, and 0 for equality.

0

-1

O	O	X
	X	X
O	X	

-1

O	O	X
O	X	X
O	X	

+1

O	O	X
	X	X
O	X	O

+1

O	O	X
X	X	X
O	X	O

-1

O	O	X
	X	
O	X	X

-1

O	O	X
O	X	
O	X	X

0

O	O	X
	X	O
O	X	X

0

O	O	X
X	X	O
O	X	X

0

O	O	X
X	X	
O	X	

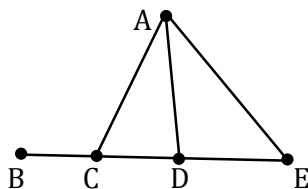
0

O	O	X
X	X	O
O	X	

+1

O	O	X
X	X	
O	X	O

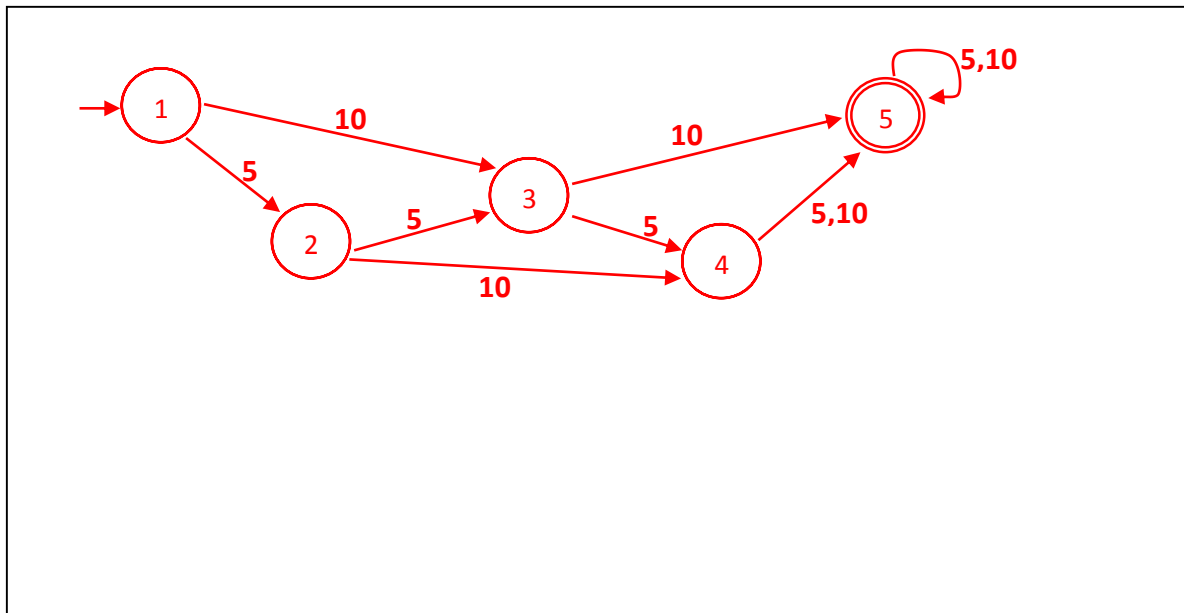
- (30P) Find the chromatic number of graph given below. Comment whether there is a Hamilton circuit or not. If there is, write it, otherwise explain the cause.



Chromatic number of the graph is 3. Here, C and E can be red, B and D can be blue, and A can be green.

There is no Hamilton circuit possibility here. Because B has only one neighbour, it can not be in any circuit. Also by edge removal, Dirac's, or Ore's theorems, we can conclude it.

3. (30P) Draw such a deterministic finite state automaton which is defined on $\{5, 10\}^*$ that it accepts only 20 as the total. You can consider each symbol as a coin.



4. (10P) For K_6 graph, we can use only 0 or 1 for the weights of edges. Prove that we can find at least one triangle in which its total weight is 3 or 0.

